

Critical relaxation and spinodal instabilities in Bayesian holographic QCD

ABSTRACT: We determine the low-lying real-time spectrum of a Bayesian-calibrated Einstein–Maxwell–dilaton model across its finite-density phase structure. Solving the coupled helicity-one and helicity-zero fluctuations, we identify the shear, sound, and baryon/heat-diffusion poles and reproduce independently computed charged-hydrodynamic coefficients at small momentum. At the maximum-likelihood critical endpoint, diffusion vanishes as the susceptibility diverges. Fits along a constant-chemical-potential approach give the dynamic exponent $z = 4.08 \pm 0.05$, where the uncertainty is dominated by the scaling window; a direct critical QNM fit gives $-\text{Im } \omega \propto q^{3.99}$. Inside the spinodal branch at $\mu_B = 650$ MeV, the longitudinal pole develops a closed unstable band. At its midpoint we find $q_* = 0.283$, maximum growth $\Gamma_*/(2\pi T) = 1.41 \times 10^{-3}$, and band edge $q_{\text{edge}} = 0.417$; the band shrinks toward either fold. A weighted 25-medoid compression of all successful posterior draws represents the full posterior weight and yields $z = 4.075^{+0.010}_{-0.013}$ at 95% credibility before the common fit-window systematic. All medoids retain negative diffusion inside the spinodal region. These results provide a validated real-time characterization of critical slowing and spinodal growth in a lattice-constrained holographic plasma.

KEYWORDS: Gauge-Gravity Correspondence, Non-Equilibrium Field Theory, Finite Temperature or Finite Density, Quark-Gluon Plasma

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1 Introduction

The real-time response of strongly interacting matter at nonzero baryon density is difficult to determine from first principles. Lattice QCD constrains equilibrium thermodynamics near vanishing chemical potential, but the sign problem obstructs direct access to the dense phase diagram and Minkowski observables. Bottom-up Einstein–Maxwell–dilaton (EMD) models offer a complementary laboratory in which a lattice-constrained equation of state, transport, and retarded response follow from the same black-brane solution.

A critical endpoint is a dynamical as well as a thermodynamic object. Conserved densities, their mode mixing, and the divergence of static susceptibilities determine critical slowing down. The QCD critical endpoint is expected to exhibit Model-H dynamics [1, 2], whereas classical large- N_c holography suppresses some nonlinear mode couplings and can realize a different effective exponent [3, 4]. The exponent and the pole responsible for it must therefore be extracted from the real-time spectrum rather than inferred from the static equation of state.

The metastable branches of a first-order transition pose a second dynamical question. Between the two spinodal boundaries a thermodynamic curvature changes sign, and a longitudinal hydrodynamic mode becomes unstable. Full gravitational dynamics must determine whether the instability occupies a finite momentum band, its fastest-growing wavelength, and the range over which a gradient expansion remains quantitative. Holographic spinodal modes have been studied in Einstein–scalar models [5–7]; the corresponding charged, lattice-calibrated EMD calculation requires the complete helicity-zero system.

The PHA analysis of ref. [8] calibrated polynomial–hyperbolic EMD models to lattice data and released the posterior ensemble [9]. Homogeneous nonhydrodynamic QNMs were previously studied in a different QCD-like EMD model [10], and recent work propagates the PHA posterior to equilibrium and horizon transport [11]. These results do not determine the coupled finite-momentum pole structure, the critical diffusion law, or the full spinodal band. Charged-fluid studies further show that mode mixing and pole collisions can sharply limit hydrodynamics [12, 13]. Recent improved-holographic-QCD, V-QCD, and Einstein–dilaton calculations further illustrate the range of finite-density instabilities and channel spectra [14–16].

This work closes that gap. We first reconstruct the horizon-data surface, continue fixed- μ_B branches through both folds, and independently locate the cusp of the thermodynamic map. We then solve the coupled helicity-one and helicity-zero perturbations in ingoing Eddington–Finkelstein coordinates. Gauge-invariant sources are evaluated on a physical horizon basis after quotienting exact residual-gauge solutions. Adaptive DOP853 shooting and an independent multidomain Chebyshev integration provide two radial methods.

On stable backgrounds, the resulting shear, sound, and baryon/heat-diffusion poles reproduce a separately constructed charged-hydrodynamic evolution matrix. Along a fixed- $\mu_B = \mu_B^c$ approach, diffusion vanishes and the susceptibility diverges; the inferred exponent is tested directly through the critical QNM dispersion. Inside the spinodal region, the full helicity-zero pole is followed beyond the range where hydrodynamics predicts unbounded q^2 growth, revealing a finite unstable band.

Finally, we propagate uncertainty through 25 weighted posterior medoids. The medoids are selected in joint parameter/critical-point space, include the maximum-likelihood draw, and represent all 1589 posterior samples whose released critical-point calculation succeeded. For every medoid we independently reconstruct the cusp, approach it along a physical path, and locate both spinodals at $\mu_B^c + 50$ MeV. We keep numerical, scaling-window, and posterior uncertainties separate.

Section 2 defines the calibrated model and phase structure. Sections 3 and 4 give the coupled systems and validation. Stable hydrodynamics, critical dynamics, spinodal growth, and posterior propagation are presented in sections 5–8. Appendices collect symbolic identities, the error budget, background and thermodynamic reproduction, homogeneous trajectories and the 2018-model benchmark, and complete reproducibility details.

2 PHA EMD model and phase structure

We use natural units, mostly-plus signature, and set the asymptotic AdS radius to one. The action is

$$S = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} \left[R - \frac{1}{2}(\partial\phi)^2 - V(\phi) - \frac{1}{4}f(\phi)F_{MN}F^{MN} \right]. \quad (2.1)$$

The polynomial–hyperbolic parametrization is

$$\begin{aligned} V(\phi) &= -12 \cosh(\gamma\phi) + b_2\phi^2 + b_4\phi^4 + b_6\phi^6, \\ f(\phi) &= \frac{\text{sech } g(\phi) + d_1 \text{sech}(d_2\phi)}{1 + d_1}, \quad g(\phi) = c_1\phi + c_2\phi^2 + c_3\phi^3. \end{aligned} \quad (2.2)$$

The maximum-likelihood realization is sample 74 of the public posterior. Its exact binary64 parameters are stored in the machine-readable configuration; the rounded values most relevant here are

$$\begin{aligned} \Lambda &= 1123.870 \text{ MeV}, & \kappa_5^2 &= 11.37047, & \gamma &= 0.590424, \\ (b_2, b_4, b_6) &= (0.327227, -0.0480134, 0.000938515). \end{aligned} \quad (2.3)$$

The Maxwell parameters are $(c_1, c_2, c_3) = (0.0088005, 0.155799, 0.0528185)$ and $(d_1, d_2) = (1.71775, 1679.18)$. The large d_2 creates a narrow UV structure in $f(\phi)$ that is retained in all backgrounds and perturbation kernels.

In domain-wall gauge,

$$ds^2 = e^{2A(r)}[-h(r)dt^2 + d\mathbf{x}^2] + \frac{dr^2}{h(r)}, \quad A_M dx^M = \Phi(r)dt. \quad (2.4)$$

We choose $r_H = A_H = h_H = \Phi_H = 0$ and $h'_H = 1$. The independent horizon data are (ϕ_0, Φ_1) , with $|\Phi_1| < [-2V(\phi_0)/f(\phi_0)]^{1/2}$. The conserved flux $\mathcal{Q} = e^{2A}f\Phi'$ is evolved analytically, and the UV coefficients give

$$\begin{aligned} T &= \frac{\Lambda}{4\pi\phi_A^{1/\nu}\sqrt{h_0}}, & \mu_B &= \frac{\Phi_0\Lambda}{\phi_A^{1/\nu}\sqrt{h_0}}, \\ s &= \frac{2\pi\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}}, & \rho_B &= -\frac{\Phi_2\Lambda^3}{\kappa_5^2\phi_A^{3/\nu}\sqrt{h_0}}, \end{aligned} \quad (2.5)$$

where $\nu = 4 - \Delta$ and $\Delta = 2 + \sqrt{4 + m^2}$.

2.1 Cusp, coexistence, and spinodals

Continuation is performed in horizon data rather than by isolated inversion in (T, μ_B) . Folds satisfy

$$\det \frac{\partial(\log T, \log \mu_B)}{\partial(\phi_0, \Phi_1/\Phi_1^{\max})} = 0. \quad (2.6)$$

At the cusp, the derivative of this determinant along the null direction of the same Jacobian also vanishes. Along fixed chemical potential, pressure differences follow from $dP = s dT$; equal pressure selects coexistence.

The 141×121 MAP horizon surface resolves 65 fixed- μ_B lines from 594 to 850 MeV. Grid coarsening changes spinodal temperatures by at most 0.018 MeV and coexistence temperatures by at most 0.0085 MeV. The independent cusp is

$$(T_c, \mu_B^c) = (103.7553, 593.3227) \text{ MeV}, \quad (\phi_0, \Phi_1/\Phi_1^{\max}) = (3.528281, 0.346374). \quad (2.7)$$

An independent 800-line crossing construction gives (103.7611, 593.3845) MeV. The HDF5 metadata for sample 74 instead store (103.8978, 602.4749) MeV. Exact parameters and the reference UV prescription were used, so the 9.152 MeV chemical-potential offset is retained as a provenance discrepancy; the local cusp in eq. (2.7) defines the dynamical background.

Figure 1 shows the resulting branch-resolved phase diagram. Stable, metastable, and unstable states used below are selected on these fixed- μ_B continuations, not on a fixed-horizon scan.

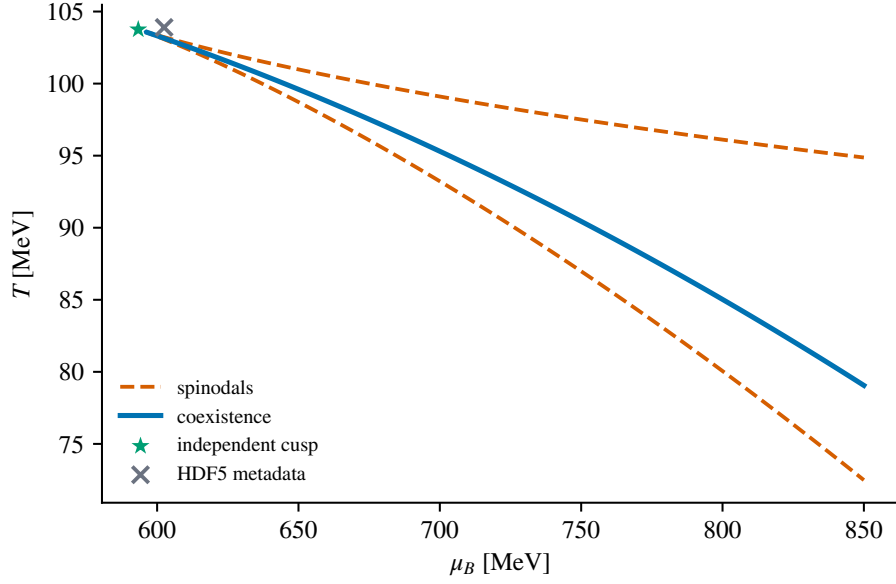


Figure 1. MAP phase structure. Orange dashed curves are the two spinodals, the blue curve is equal-pressure coexistence, and the green star is the independently located cusp. The gray cross marks the distinct critical coordinates stored in the HDF5 metadata.

3 Coupled perturbations and charged hydrodynamics

We transform eq. (2.4) to ingoing Eddington–Finkelstein coordinates,

$$ds^2 = -e^{2A}h dv^2 + 2e^A dv dr + e^{2A}d\mathbf{x}^2, \quad A = \Phi(r)dv, \quad (3.1)$$

and use $e^{-i\omega v + ikz}$. Stable poles have $\text{Im } \omega < 0$. We report $\mathfrak{w} = \omega/(2\pi T)$ and $\mathfrak{q} = k/(2\pi T)$; the horizon-normalized solver uses $\mathfrak{w} = 2\omega_{\text{num}}$ and $k_{\text{num}} = \mathfrak{q}/(2\sqrt{h_0})$.

At finite momentum the fluctuations split into $SO(2)$ helicities. Helicity two is the decoupled tensor field. The helicity-one primitives are (H_{vx}, H_{zx}, a_x) , while helicity zero contains

$$(H_{vv}, H_{vz}, H_{zz}, H_{aa}, a_v, a_z, \varphi), \quad H_{aa} = H_{xx} = H_{yy}. \quad (3.2)$$

The complete linearized Einstein, Maxwell, and scalar equations are generated from eq. (2.1). They remain linear in $\sigma = -i\omega$ and use $p = ik$.

3.1 Gauge invariants and source matrix

In helicity one the physical combinations are

$$Z_{\text{sh}} = pH_{vx} - \sigma H_{zx}, \quad Z_J = a_x. \quad (3.3)$$

For helicity zero we use

$$\begin{aligned} Z_H &= p^2 H_{vv} - 2p\sigma H_{vz} + \sigma^2 H_{zz} + \left[p^2 \left(h + \frac{h'}{2A'} \right) - \sigma^2 \right] H_{aa}, \\ Z_A &= pa_v - \sigma a_z - \frac{p\Phi'}{2A'} H_{aa}, \quad Z_\phi = \varphi - \frac{\phi'}{2A'} H_{aa}. \end{aligned} \quad (3.4)$$

These combinations annihilate the exact residual radial-gauge orbits. At a trial frequency, regular horizon data span seven and three solutions in helicities zero and one, respectively. Quotienting four and one analytic gauge solutions leaves three and two physical solutions. Their UV source coefficients form square matrices $S_0(\omega, k)$ and $S_1(\omega, k)$; QNMs satisfy

$$\det S_s(\omega, k) = 0, \quad (3.5)$$

or, equivalently, a vanishing smallest singular value. This is the standard coupled-field source-matrix construction [17, 18].

Metric, Maxwell, and scalar source/vev powers are respectively u^0/u^4 , u^0/u^2 , and u^ν/u^Δ for $u = e^{-A}$. We extract each source by a UV value/derivative combination rather than fitting an exponentially small difference. Primitive radial Einstein and Maxwell equations not used for evolution are evaluated as constraints on the physical source-null solution.

3.2 Independent hydrodynamic prediction

Local derivatives of (T, μ_B, s, ρ_B) with respect to the two horizon data are transformed to derivatives in (ϵ, ρ_B) . The Maxwell relation $\partial s / \partial \mu_B|_T = \partial \rho_B / \partial T|_{\mu_B}$ provides an independent check. We use $\eta = s/(4\pi)$ and the charged Eling–Oz horizon formula [19]

$$\frac{\zeta}{\eta} = \left(s \frac{\partial \phi_H}{\partial s} + \rho_B \frac{\partial \phi_H}{\partial \rho_B} \right)^2. \quad (3.6)$$

The finite incoherent conductivity of the translationally invariant EMD plasma is [20]

$$\frac{\sigma_Q}{T} = \frac{2\pi\sqrt{h_0} f(\phi_H)}{\kappa_5^2} \left(\frac{Ts}{\epsilon + P} \right)^2. \quad (3.7)$$

Rather than using $D_B = \sigma/\chi$ outside its neutral limit, we diagonalize the full linearized charged-fluid evolution matrix for $(\delta\epsilon, \delta\rho_B, \delta u_z)$. It gives

$$\begin{aligned} \mathfrak{w}_{\text{sh}} &= -iD_\eta \mathfrak{q}^2 + \mathcal{O}(\mathfrak{q}^4), \\ \mathfrak{w}_\pm &= \pm c_s \mathfrak{q} - i\Gamma_s \mathfrak{q}^2 + \mathcal{O}(\mathfrak{q}^3), \\ \mathfrak{w}_D &= -iD_B \mathfrak{q}^2 + \mathcal{O}(\mathfrak{q}^4), \end{aligned} \quad (3.8)$$

including charge/heat mixing. All coefficients compared with QNMs below are computed from equilibrium and horizon data before any dispersion fit.

4 Numerical method and validation

Backgrounds are integrated from a second-order regular horizon series with DOP853, relative tolerance 3×10^{-12} , absolute tolerance 3×10^{-14} , and maximum step 0.025. The conserved electric flux is used directly, and UV coefficients are extracted on a fixed sampling grid. Across the production phase surface, normalized background constraints stay below 4.8×10^{-9} and Gauss drift below 9.0×10^{-8} .

For coupled QNMs, the characteristic horizon equations and the radial Maxwell constraint determine the regular first derivative. Differentiating the six characteristic helicity-zero equations once and adding the noncharacteristic Maxwell equation fixes the second horizon coefficient. Including this term is essential: a first-order start excites a constraint-violating DAE solution, whereas the second-order recurrence makes primitive constraints converge with the horizon start.

The primary source-matrix solver integrates the physical horizon basis with DOP853 from $r_0 = 10^{-6}$ to $r_{UV} = 5$. The independent method converts the same second-order system to first order and solves it on eight Chebyshev subdomains separated at

$$r = (10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1.5, 3), \quad (4.1)$$

using 24 intervals per domain. It neither calls an ODE integrator nor forms the global primitive QZ pencil. Resolution, UV radius, horizon start, and the subdomain partition are varied independently.

The six benchmarks in table 1 comprise shear, sound, and diffusion on a neutral background and on a charged stable background. The deliberately coarse $r_0 = 2 \times 10^{-6}$ variant has the largest constraint; halving r_0 lowers it while changing $\mathfrak{w}$ by only $\mathcal{O}(10^{-11})$. The MAP critical diffusion pole and central spinodal maximum are also recomputed by shooting and by a 28-interval refined collocation. Their respective method distances are 3.1×10^{-11} and 1.5×10^{-10} .

The rejected global primitive helicity-zero QZ pencil has eigenvalue condition numbers as large as 10^{15} and does not satisfy continuum constraints. It is not used for a frequency, an uncertainty, or a claim. This exclusion is important for a non-normal problem in which small discrete residuals alone do not guarantee spectral stability [21, 22].

5 Stable-phase hydrodynamic poles

We first test the coupled calculation away from criticality. Eight momenta $0.02 \leq \mathfrak{q} \leq 0.12$ are used on a neutral state at $T = 282.77$ MeV and a charged state at $(T, \mu_B) = (174.245, 348.490)$ MeV. Fits include the first subleading power in $\mathfrak{q}$ and extract the $\mathfrak{q} \rightarrow 0$ coefficient.

The agreement in table 2 identifies all three poles without using field-content heuristics. In particular, the longitudinal diffusive root is the charge/heat eigenmode of the complete thermodynamic matrix. The largest primitive constraint in the 48 roots is 9.1×10^{-6} and the largest source determinant is 5.6×10^{-12} .

6 Critical slowing at the endpoint

We approach the MAP cusp on the hot stable branch at fixed $\mu_B = \mu_B^c$. The horizon coordinate is displaced by $0.5 \geq \phi_0^c - \phi_0 \geq 0.005$, and the electric datum is solved anew at every point to keep μ_B fixed. Define the dimensionless distance

$$r = \left[\left(\frac{T - T_c}{T_c} \right)^2 + \left(\frac{\mu_B - \mu_B^c}{\mu_B^c} \right)^2 \right]^{1/2}. \quad (6.1)$$

Table 1. Acceptance summary for six neutral/charged hydrodynamic benchmark poles. The continuum envelope includes $N = 20, 24, 28$ per subdomain, $r_0 = (0.5, 1, 2) \times 10^{-6}$, $r_{UV} = 4.5, 5.0, 5.5$, and an independent partition.

diagnostic	maximum or envelope
symbolic EMD/gauge/constraint tests	10/10 passed
horizon residual-gauge defect	4.1×10^{-16}
normalized source determinant	1.6×10^{-12}
primitive constraint, nominal	9.0×10^{-6}
primitive constraint, full r_0 study	2.41×10^{-5}
shooting-collocation distance in $\mathfrak{w}$	3.7×10^{-12}
continuum envelope in $\mathfrak{w}$	3.7×10^{-11}
hydro/QNM difference at $\mathfrak{q} = 0.05$	1.7×10^{-3} relative

Table 2. Independent hydrodynamic coefficients and their QNM small-momentum fits. The sound attenuation is defined by $\text{Im } \mathfrak{w}_+ = -\Gamma_s \mathfrak{q}^2 + \dots$.

state	coefficient	horizon/EOS	QNM fit
neutral	D_η	0.500000	0.499998
	c_s	0.532966	0.532966
	D_B	0.998757	0.998671
	Γ_s	0.375932	0.375933
charged	D_η	0.458120	0.458119
	c_s	0.448585	0.448585
	D_B	0.736939	0.736949
	Γ_s	0.392189	0.392189

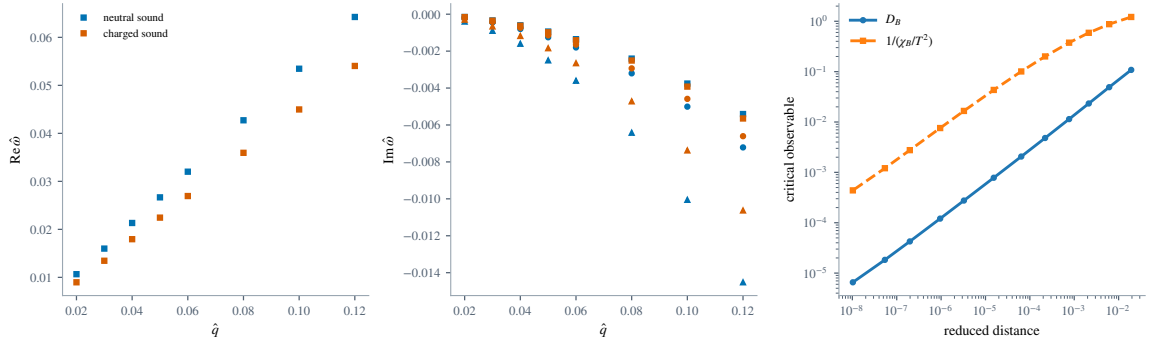


Figure 2. Stable and critical hydrodynamics. (a) Positive sound dispersion on the neutral and charged backgrounds. (b) Damping of shear (circles), sound (squares), and diffusion (triangles); symbols are full QNMs and the small- $\mathfrak{q}$ behavior is fixed by the independently constructed hydrodynamic matrix. (c) Along $\mu_B = \mu_B^c$, D_B vanishes while $(\chi_B/T^2)^{-1}$ approaches zero. Lines connect computed points and are not additional fits.

Over the asymptotic window, $D_B \sim r^{0.64--0.66}$ and $\chi_B/T^2 \sim r^{-0.62--0.63}$. With the classical holographic static anomalous dimension $\eta_{\text{stat}} = 0$, dynamic scaling gives

$$D_B \sim \xi^{2-z}, \quad \chi_B \sim \xi^2, \quad z = 2 - 2 \frac{d \log D_B}{d \log \chi_B}. \quad (6.2)$$

Varying the fit over the last four through eight points gives

$$z = 4.08 \pm 0.05 \quad (\text{fit-window systematic}). \quad (6.3)$$

The local thermodynamic Jacobian condition number rises from 1.3×10^2 to 2.8×10^6 along this path, while the Maxwell-relation residual stays below 3.7×10^{-6} .

At the cusp itself, the q^2 coefficient vanishes and the full QNM determines the leading gradient power. Seven momenta from $\mathbf{q} = 0.04$ to 0.15 give

$$-\text{Im } \mathfrak{w}_D = A \mathbf{q}^z, \quad z = 3.99 \quad (0.04 \leq \mathbf{q} \leq 0.08), \quad (6.4)$$

with local exponents decreasing from 3.996 to 3.919 as the upper end is moved to $\mathbf{q} = 0.15$. We quote $z = 3.99(2)$ for the direct dispersion, where the parenthesis covers this controlled window drift. The largest critical-QNM constraint is 9.1×10^{-7} and the source determinant is below 1.7×10^{-11} .

The homogeneous modes do not close at the cusp. The least-damped gapped triplet has $\mathfrak{w} = 1.68854 - 0.10317i$, while the leading singlet and tensor dampings are 0.2561 and 0.8669. Critical slowing is therefore carried by the conserved longitudinal mode, not by an accidental homogeneous gapped pole. This distinction is consistent with earlier holographic studies of nonhydrodynamic critical behavior [3, 23].

7 Spinodal growth and hydrodynamic breakdown

At $\mu_B = 650$ MeV the MAP fixed-chemical-potential line has folds at $\phi_0 = 3.18138$ and 3.88392. We solve the longitudinal spectrum at fractional positions 0.1, 0.5, and 0.9 between them. In all cases the signed hydrodynamic diffusion coefficient is negative, so $\text{Im } \mathfrak{w}_D = -D_B \mathbf{q}^2 > 0$ at small momentum.

Table 3 demonstrates a closed unstable band rather than only a thermodynamic sign change. The central maximum occurs at $\mathbf{q}_* = 0.2826$ with $\Gamma_* = 0.001414$ and the pole returns to the lower half plane at $\mathbf{q}_{\text{edge}} = 0.4175$. Shooting at $\mathbf{q} = 0.28$ agrees with nominal collocation within 1.5×10^{-10} in $\mathfrak{w}$; the refined collocation agrees to displayed precision and has primitive constraint 3.3×10^{-7} .

The q^2 hydrodynamic approximation correctly gives the initial slope but cannot close the band. Its 10% range ends at $\mathbf{q} = 0.05\text{--}0.08$, well below both the fastest-growing scale and the edge. The maximum growth, preferred momentum, and band edge all decrease toward either fold. Three positions are enough to establish this trend but not to claim a spinodal-boundary exponent; we therefore do not fit one.

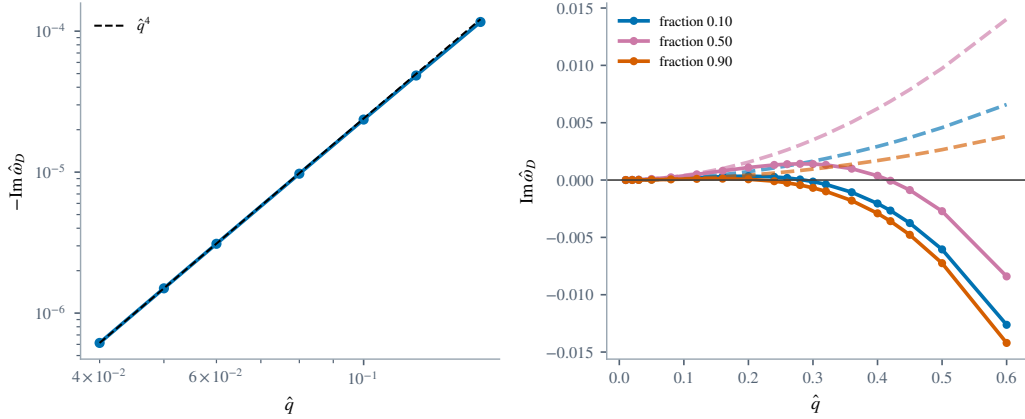


Figure 3. Critical and spinodal longitudinal dynamics. (a) Diffusion at the MAP cusp follows $-\text{Im } w_D \sim \hat{q}^4$ at small momentum. (b) Full-QNM growth rates at three fractional positions between the $\mu_B = 650$ MeV folds. Dashed curves are the q^2 hydrodynamic prediction; full QNMs form a closed band and become damped at finite momentum.

Table 3. Full-QNM unstable bands at $\mu_B = 650$ MeV. The fractional position runs from the hot to the cold fold. $q_{10\%}$ is the largest sampled momentum where the hydrodynamic growth rate agrees with the full QNM within 10%.

fraction	T [MeV]	D_B	q_*	Γ_*	q_{edge}	$q_{10\%}$
0.10	98.786	-0.01829	0.198	3.41×10^{-4}	0.286	0.05
0.50	99.940	-0.03895	0.283	1.414×10^{-3}	0.417	0.08
0.90	100.924	-0.01060	0.154	1.22×10^{-4}	0.221	0.05

8 Posterior robustness

The HDF5 posterior contains 1600 draws, of which 1589 have successfully released critical-point metadata. Because these are already posterior draws, their base weights are equal. We standardize the 11 model parameters together with the stored critical coordinates, retain principal components carrying 95% of the variance, and select 25 deterministic cluster medoids. Each medoid is weighted by its Voronoi population, and sample 74 is forced into the ensemble. The resulting weights sum to one and represent all 1589 successful draws.

For each medoid, the cusp is independently located from the fold and null-direction conditions rather than copied from metadata. Adaptive finite-difference steps from $(0.01, 0.001)$ through $(0.03, 0.003)$ are used only to control the nearly singular map; the maximum accepted cusp-equation residual is 1.8×10^{-4} . Every medoid then follows the fixed- μ_B^c critical path and a fixed- $\mu_B^c + 50$ MeV line through both folds.

Figure 4 and table 4 show that the posterior spread of the exponent is smaller than the scaling-window systematic. We therefore summarize the ensemble as $z = 4.08 \pm 0.05$, not with three-decimal precision. The wider posterior interval of the distance exponent reflects sensitivity to how the physical path approaches the local scaling axes; the D_B - χ_B relation and direct MAP q^4 dispersion are more stable. All 25 medoids possess two folds and have

Table 4. Weighted 25-medoid posterior summary. Intervals are the 2.5th and 97.5th percentiles of the represented posterior. The separate common z fit-window systematic is approximately ± 0.05 .

observable	2.5%	median	97.5%
T_c [MeV]	101.48	103.88	105.75
μ_B^c [MeV]	572.43	591.59	622.48
z from D_B - χ_B scaling	4.062	4.075	4.085
D_B distance exponent	0.457	0.648	0.676
ΔT_{sp} [MeV] at $\mu_B^c + 50$ MeV	1.843	1.876	1.903
midpoint D_B in the spinodal	-0.03836	-0.03457	-0.03235

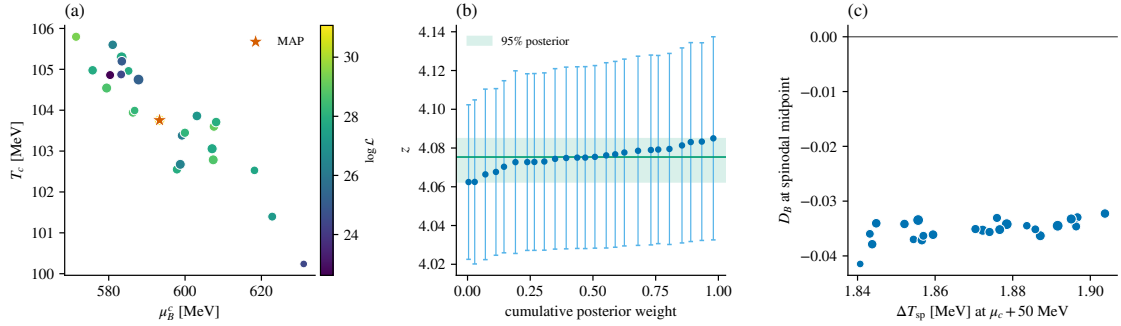


Figure 4. Controlled posterior propagation. Marker area is proportional to represented posterior weight. (a) Independently reconstructed critical endpoints, colored by log likelihood; the star is the MAP realization. (b) Medoid exponents with their fit-window half-ranges; the green band is the weighted 95% posterior interval before the common systematic. (c) Spinodal temperature width and signed midpoint diffusion at $\mu_B^c + 50$ MeV. Every medoid remains below $D_B = 0$.

negative midpoint D_B , so the existence of spinodal growth is posterior-robust. Detailed $\mathbf{q}_*$ and Γ_* values remain full-QNM results for the MAP realization; we do not present them as posterior intervals.

9 Conclusions

We have computed the coupled finite-momentum real-time spectrum of the Bayesian-calibrated PHA EMD model. Gauge-invariant source matrices, exact residual-gauge quotienting, second-order horizon recurrences, primitive constraint monitoring, and two independent radial integrations turn the helicity-one and helicity-zero systems into controlled continuum QNM calculations.

On stable backgrounds, the shear, sound, and baryon/heat-diffusion poles match transport and thermodynamic coefficients computed without QNM input. This agreement identifies the longitudinal modes and verifies the charged-fluid mixing. The maximum relative difference in the dedicated $\mathbf{q} = 0.05$ benchmarks is 1.7×10^{-3} , and extrapolated coefficients agree more closely.

At the independently located MAP critical endpoint, the diffusion coefficient vanishes while the baryon susceptibility diverges. The hydrodynamic scaling relation gives $z = 4.08 \pm 0.05$, and the full critical QNM independently obeys $-\text{Im } \mathbf{w}_D \sim \mathbf{q}^{3.99}$ at small momentum. Homogeneous gapped modes retain finite damping; the conserved diffusive pole is the critical degree of freedom.

Inside the $\mu_B = 650$ MeV spinodal branch, diffusion crosses into the upper half frequency plane and forms a closed unstable band. At the midpoint its fastest-growing scale is $\mathbf{q}_* = 0.283$, its maximum dimensionless growth is 1.41×10^{-3} , and its edge is $\mathbf{q} = 0.417$. Hydrodynamics controls only the initial q^2 segment and cannot predict closure. The band shrinks toward both spinodal boundaries.

A weighted 25-medoid posterior analysis represents all successful released draws and reconstructs every local cusp and spinodal pair independently. It gives a weighted 95% range $4.062 < z < 4.085$ before the dominant common fit-window error, and every medoid retains negative diffusion inside the spinodal. This justifies a Bayesian robustness claim for critical slowing and the existence of spinodal growth, while MAP-only band scales are labeled as such.

The interpretation remains subject to classical large N_c , strong coupling, translational invariance, equilibrium backgrounds, and the bottom-up model Ansatz. The results characterize a lattice-constrained holographic model of QCD, not QCD itself. Natural extensions include correlator residues, hydrodynamic/nonhydrodynamic pole collisions at complex momentum, nonlinear spinodal evolution, and full-QNM propagation beyond the thermodynamic medoid ensemble.

Acknowledgments

OpenAI Codex was used as an AI-assisted tool during preparation of this work for code refactoring, workflow automation, construction of numerical and symbolic validation scripts, literature-metadata checks, and language and layout editing. Its outputs were checked against the version-controlled source, machine-readable results, cited primary literature, and independent numerical tests. Responsibility for the scientific content and the submitted manuscript remains with the authors.

A Linearized identities and degree counting

The generated perturbation artifact contains three evolution equations and one primitive Einstein constraint in helicity one. Helicity zero contains seven evolution/characteristic equations and the unused primitive equations E_{rr} , E_{rv} , E_{rz} , and M_r . Direct substitution verifies the linearized Bianchi and Maxwell identities on every background.

The residual helicity-one orbit is proportional to $(H_{vx}, H_{zx}, a_x) = (-\sigma X, -pX, 0)$, so three primitives minus one gauge orbit give the two physical sources in eq. (3.3). In helicity zero the four residual functions obey

$$T' = 0, \quad R' = -A'R - \sigma T, \quad Z' = -pe^{-A}T, \quad \lambda' = 0. \quad (\text{A.1})$$

Seven regular primitives minus four gauge directions give the three sources in eq. (3.4). The validation suite also checks transverse parity, charge conjugation, the neutral limit, the $k \rightarrow 0$ triplet and singlet reductions, and all UV source/vev powers. All ten symbolic tests pass.

B Numerical error budget

The coupled validation table is stored row by row, including frequency, source determinant and singular value, gauge defect, primitive constraints, horizon-recurrence condition, local principal-matrix condition, collocation residual, r_0 , r_{UV} , resolution, and optimizer status. A root is accepted by the source residual even when MINPACK reports stagnation, provided that the independently evaluated determinant, continuum envelope, and radial cross-check pass. Optimizer status alone is never a physics criterion.

For the critical exponent, five nested windows containing the last four through eight thermodynamic points give 4.044, 4.057, 4.079, 4.107, and 4.142. Their half-range sets the quoted ± 0.05 systematic. The direct QNM fits over four through seven low-momentum points give exponents 3.987, 3.982, 3.976, and 3.968. Posterior and fit-window uncertainties are reported separately because they arise from different sources.

C Background, CEP, and thermodynamic reproduction

C.1 Exact kernels and background equations

All model derivatives are evaluated analytically. In particular,

$$\begin{aligned} V_\phi &= -12\gamma \sinh(\gamma\phi) + 2b_2\phi + 4b_4\phi^3 + 6b_6\phi^5, \\ V_{\phi\phi} &= -12\gamma^2 \cosh(\gamma\phi) + 2b_2 + 12b_4\phi^2 + 30b_6\phi^4, \end{aligned} \quad (C.1)$$

$$\begin{aligned} f_{\phi\phi} &= \frac{1}{1+d_1} \left\{ \operatorname{sech} g [\tanh^2 g - \operatorname{sech}^2 g] (g')^2 - \operatorname{sech} g \tanh g g'' \right. \\ &\quad \left. + d_1 d_2^2 \operatorname{sech}(d_2\phi) [\tanh^2(d_2\phi) - \operatorname{sech}^2(d_2\phi)] \right\}. \end{aligned} \quad (C.2)$$

Figure 5 shows the exact MAP kernels. The narrow structure in $f(\phi)$ near the boundary, controlled by the large value of d_2 , is resolved rather than smoothed or removed.

For the background ansatz in eq. (2.4), the equations used by the independent reference integrator are

$$\begin{aligned} \phi'' + \left(4A' + \frac{h'}{h}\right) \phi' - \frac{1}{h} \left(V_\phi - \frac{1}{2}e^{-2A}f_\phi\Phi'^2\right) &= 0, \\ \Phi'' + \left(2A' + \frac{f_\phi}{f}\phi'\right) \Phi' &= 0, \\ A'' + \frac{1}{6}\phi'^2 &= 0, \quad h'' + 4A'h' - e^{-2A}f\Phi'^2 = 0. \end{aligned} \quad (C.3)$$

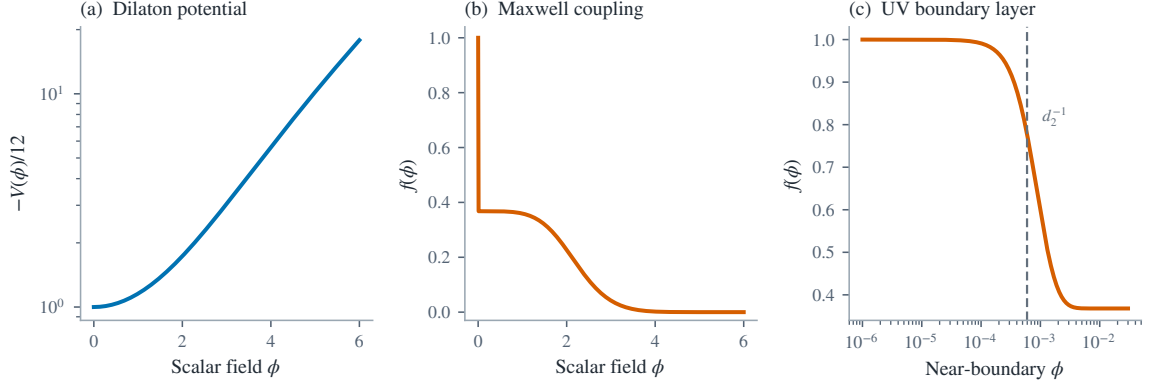


Figure 5. Exact MAP dilaton potential and Maxwell coupling. The right panel resolves the ultraviolet boundary layer at $\phi \sim d_2^{-1}$.

Table 5. Maximum-likelihood PHA realization. Exact binary64 values and provenance hashes are retained in the machine-readable configuration.

quantity	value
Λ [MeV]	1123.87
κ_5^2	11.3705
γ	0.590424
b_2	0.327227
b_4	-0.0480134
b_6	0.000938515
c_1	0.00880048
c_2	0.155799
c_3	0.0528185
d_1	1.71775
d_2	1679.18
m^2	-3.52875
Δ	2.68648

They obey the constraint and conserved charge

$$\begin{aligned} \mathcal{C} &= h(24A'^2 - \phi'^2) + 6A'h' + 2V + e^{-2A}f\Phi'^2 = 0, \\ \mathcal{Q} &= e^{2A}f\Phi' = \text{constant}. \end{aligned} \tag{C.4}$$

The reduced production path evolves $\mathcal{Q}$ directly and reconstructs

$$\Phi'(r) = \frac{\mathcal{Q}e^{-2A(r)}}{f[\phi(r)]}. \tag{C.5}$$

C.2 Regular horizon series and background diagnostics

Writing

$$\begin{aligned} A &= A_1 r + A_2 r^2 + \dots, & h &= r + h_2 r^2 + \dots, \\ \phi &= \phi_0 + \phi_1 r + \phi_2 r^2 + \dots, & \Phi &= \Phi_1 r + \Phi_2 r^2 + \dots, \end{aligned} \quad (\text{C.6})$$

the compiled recurrence reproduces

$$\begin{aligned} A_1 &= -\frac{2V_0 + f_0 \Phi_1^2}{6}, & \phi_1 &= V_{\phi,0} - \frac{1}{2} f_{\phi,0} \Phi_1^2, \\ A_2 &= -\frac{\phi_1^2}{12}, & h_2 &= \frac{1}{2} (f_0 \Phi_1^2 - 4A_1), \\ \Phi_2 &= -\frac{\Phi_1}{2} \left(2A_1 + \frac{f_{\phi,0}}{f_0} \phi_1 \right). \end{aligned} \quad (\text{C.7})$$

Higher coefficients through fourth order are generated by truncated-series arithmetic applied to the regular equations, rather than hand-entered formulae.

The conserved-charge background system is integrated from $\epsilon_H = 10^{-7}$ to $r_{\max} = 12$ with DOP853, relative tolerance 3×10^{-12} , absolute tolerance 3×10^{-14} , and maximum step 0.025. UV coefficients are extracted on a fixed 2400-point grid. Across the 90-point validation scan, the normalized Einstein constraint is at most 3.50×10^{-9} and Gauss drift is below 4.45×10^{-16} ; at the MAP cusp they are 1.88×10^{-11} and 3.33×10^{-16} , respectively.

For provenance, figure 7 also retains the direct image of the old 90-background horizon-data survey. Its charged curves are not equilibrium trajectories: folds in the map make them multivalued in thermodynamic coordinates. All Route-A physics instead uses the continued branches of figure 1.

C.3 Independent critical-point reproduction

The public reference-compatible construction follows crossings of neighboring constant- ϕ_0 curves. Increasing the number of trajectories from 100 to 800 moves the estimate from (103.666, 594.761) MeV to (103.761, 593.384) MeV. The independent local method uses

$$J = \frac{\partial(\log T, \log \mu_B)}{\partial(\phi_0, \Phi_1/\Phi_1^{\max})} \quad (\text{C.8})$$

and imposes both $\det J = 0$ and a vanishing directional derivative along the null vector of J . Table 6 makes the provenance discrepancy explicit.

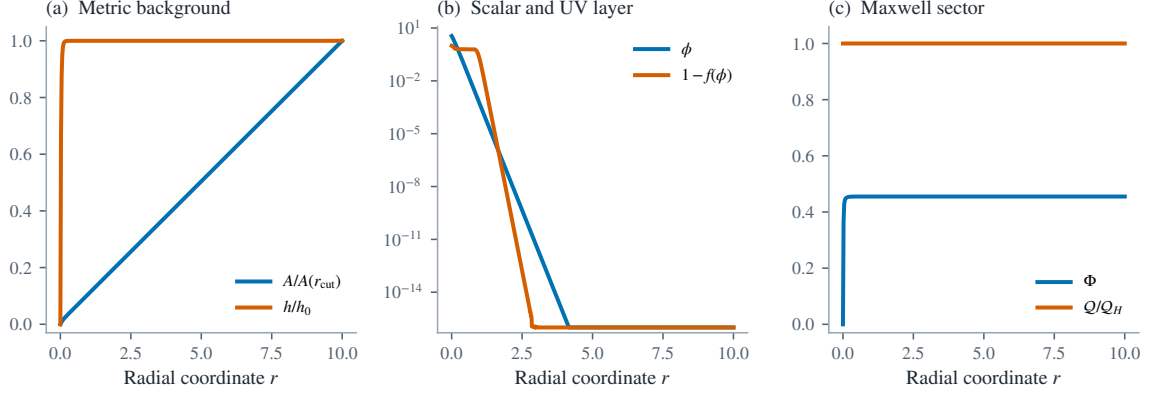


Figure 6. MAP critical background used by the homogeneous and coupled benchmarks. The middle panel resolves both the scalar decay and the narrow Maxwell-coupling layer; the right panel verifies constant electric flux.

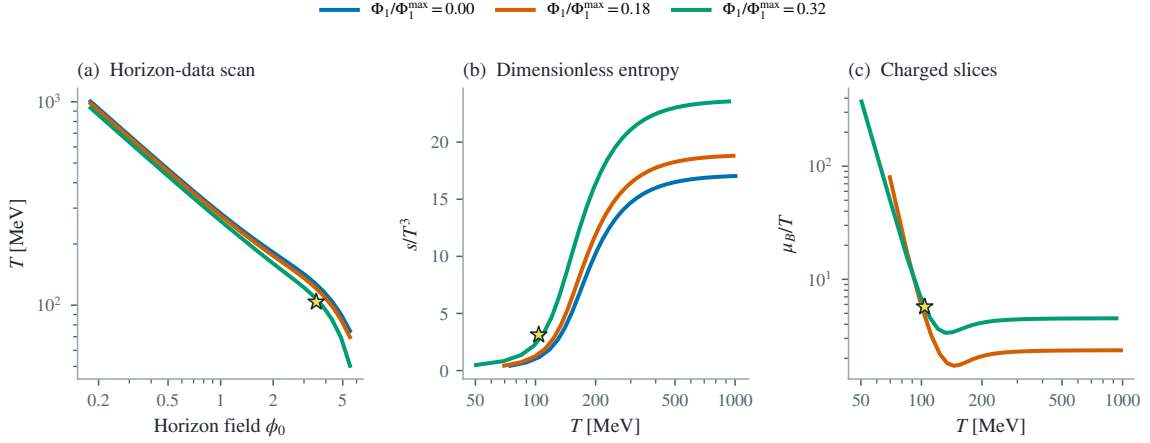


Figure 7. Exploratory horizon-data survey retained as a solver diagnostic. The nonmonotonic charged slices explain why isolated inversion in (T, μ_B) is ill-conditioned; they are not used as physical temperature scans.

Table 6. Critical coordinates from posterior metadata and two independent local reconstructions. The two numerical routes agree with one another; the distinct metadata value is not used to choose the dynamical background.

method	T_c [MeV]	μ_B^c [MeV]
HDF5 metadata	103.8978	602.4749
constant- ϕ_0 crossings ($N = 800$)	103.7611	593.3845
local cusp equations	103.7553	593.3227

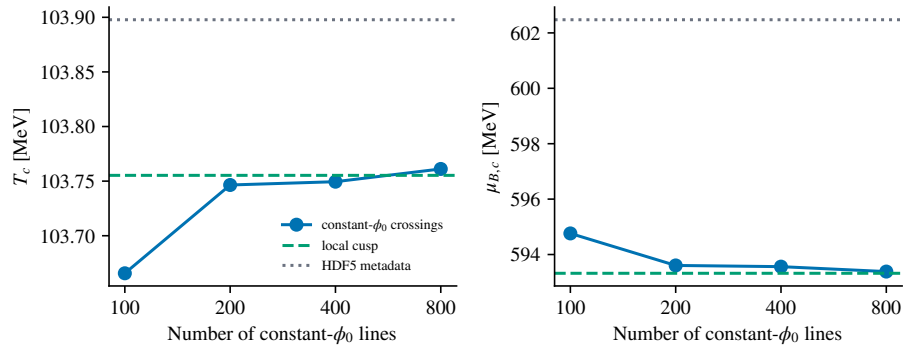


Figure 8. Convergence of the reference-compatible constant- ϕ_0 crossing algorithm. The independent local cusp agrees with the converged crossing estimate, while the HDF5 metadata remain offset.

C.4 Neutral equation of state and baryon susceptibility

As a background test independent of every QNM solver, we reproduce the neutral entropy and the second-order baryon susceptibility

$$\chi_2^B(T) = \left. \frac{\partial(\rho_B/T^3)}{\partial(\mu_B/T)} \right|_{\mu_B=0}. \quad (\text{C.9})$$

For each of 30 neutral backgrounds, the centered derivative is

$$\chi_2^B(\delta) = \frac{(\rho_B/T^3)_{+\delta} - (\rho_B/T^3)_{-\delta}}{(\mu_B/T)_{+\delta} - (\mu_B/T)_{-\delta}}, \quad \delta = 0.005, 0.010, 0.020. \quad (\text{C.10})$$

Charge conjugation makes T and s even and μ_B and ρ_B odd in δ , so fixing ϕ_0 differs from fixing T only at $\mathcal{O}(\delta^2)$. The 90 evaluations span $74.69 \leq T/\text{MeV} \leq 1003.70$; the largest susceptibility displacement relative to the finest step is 2.06×10^{-3} and occurs at the coarsest step. No smoothing or fitted interpolation enters the displayed points.

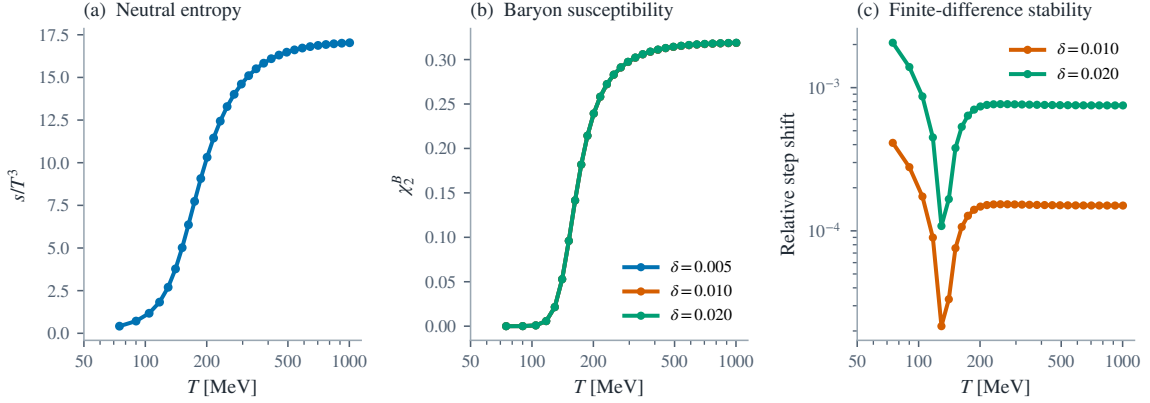


Figure 9. Reproduction and step-size stability of neutral thermodynamics. Left: s/T^3 . Center: χ_2^B from eq. (C.10). Right: relative displacement from the finest step. This equilibrium validation is independent of the finite-density critical and spinodal claims.

D Homogeneous gapped spectrum and legacy benchmark

D.1 Homogeneous master equations

At $k = 0$, $SO(3)$ organizes the perturbations into quintuplet, triplet, and singlet sectors. The decoupled tensor equation (and finite- k helicity-two benchmark) is

$$Z_2'' + \left(4A' + \frac{h'}{h} - \frac{2i\omega e^{-A}}{h} \right) Z_2' + \left(-\frac{e^{-2A}k^2}{h} - \frac{3i\omega e^{-A}A'}{h} \right) Z_2 = 0. \quad (\text{D.1})$$

The homogeneous vector master field obeys

$$a'' + \left(2A' + \frac{h'}{h} + \frac{f_\phi}{f}\phi' - \frac{2i\omega e^{-A}}{h} \right) a' + \left[-\frac{e^{-2A}f\Phi'^2}{h} - \frac{i\omega e^{-A}}{h} \left(A' + \frac{f_\phi}{f}\phi' \right) \right] a = 0. \quad (\text{D.2})$$

Their normalizable powers are $Z_2 \sim u^4$ and $a \sim u^2$ for $u = e^{-A}$. For the singlet, define

$$\mathcal{S} = \varphi - \frac{\phi'}{2A'} \frac{H_{xx} + H_{yy} + H_{zz}}{3}, \quad Y = \frac{\phi'}{A'}, \quad \psi = \frac{\mathcal{S}}{Y}. \quad (\text{D.3})$$

The source-free homogeneous master equation is

$$\begin{aligned} \psi'' + \left(4A' + \frac{h'}{h} + 2\frac{Y'}{Y} - \frac{2i\omega e^{-A}}{h} \right) \psi' \\ + \left[-\frac{h'}{h} \frac{Y'}{Y} + \frac{e^{-2A}\Phi'^2}{h\phi'} (3A'f_\phi - f\phi') - \frac{i\omega e^{-A}}{h} \left(3A' + 2\frac{Y'}{Y} \right) \right] \psi = 0. \end{aligned} \quad (\text{D.4})$$

The first-order symbolic variation of the complete EMD system reproduces eqs. (D.2) and (D.4). After UV source factoring, the collocation problem is the frequency-linear pencil

$$[M_0(k) + \omega M_1(k)]\mathbf{v} = 0, \quad (\text{D.5})$$

with the numerical conversion

$$\mathfrak{w} = 2\omega_{\text{num}}, \quad k_{\text{num}} = \frac{\mathfrak{q}}{2\sqrt{h_0^{\text{far}}}}. \quad (\text{D.6})$$

D.2 Continuum validation at the MAP cusp

The homogeneous quintuplet, triplet, and singlet operators factor the UV sources analytically. Resolution and domain errors are varied separately: $N = 64, \dots, 160$ at $r_{\text{cut}} = 10$, and $r_{\text{cut}} = 8, \dots, 12$ at $N = 160$. Right-eigenfunction overlaps track each branch, while left and right null vectors give

$$\kappa_\omega = \frac{\sqrt{1 + |\omega|^2} \|u\| \|v\|}{|u^\dagger M_1 v|}. \quad (\text{D.7})$$

The largest condition estimate among accepted modes is 2.1×10^5 . An independent complex shooting solver varies both the horizon start and UV extraction radius; no inverse of the generalized pencil is formed [24].

The leading triplet is oscillatory and gapped; its small damping is not a baryon-diffusion pole. Three former discrete-pencil roots fail the independent source test and are excluded. Figure 10 retains the full continuum evidence for the four accepted modes.

D.3 Controlled thermodynamic trajectories

The leading pole in each sector is continued through 18 backgrounds on each of the physical curves $\mu_B = 0$ and $\mu_B/T = 2$, spanning approximately $87 \leq T \leq 346$ MeV. Every point is a shooting root; the associated factored-collocation eigenfunction is assigned using frequency and overlap with its parent. The minimum parent overlap is 0.9985 and the largest pointwise shooting/spectral difference is 0.012 in $\mathfrak{w}$.

The quintuplet and singlet damping rates are nonmonotonic but remain of order unity or a few tenths. The leading triplet changes more strongly: toward the low-temperature endpoint, $-\text{Im } \mathfrak{w}$ falls to 3.7×10^{-3} while the real part remains nonzero. A dedicated endpoint study through $N = 160$ reproduces this damping within 2×10^{-7} .

Table 7. Independently validated homogeneous poles at the MAP cusp. Parentheses give the conservative spectral/shooting envelope on the last displayed digits.

sector	n	$\mathfrak{w}_n$	source residual	κ_ω
quintuplet	0	$1.8732(3) - 0.8669(6)i$	9.5×10^{-15}	2.1×10^5
triplet	0	$1.688542(4) - 0.103168(6)i$	3.9×10^{-14}	4.6×10^2
triplet	1	$2.8976(34) - 0.8634(12)i$	9.6×10^{-14}	1.4×10^4
singlet	0	$1.4924(12) - 0.2561(13)i$	9.3×10^{-12}	5.1×10^2

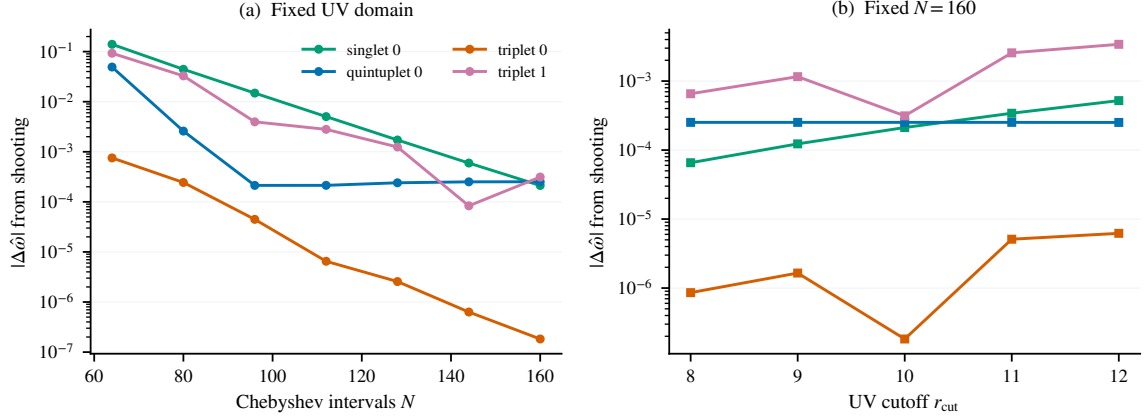


Figure 10. Continuum checks for the four accepted homogeneous poles. Distances are measured from the complex-shooting root; radial resolution and UV-domain variations are separated.

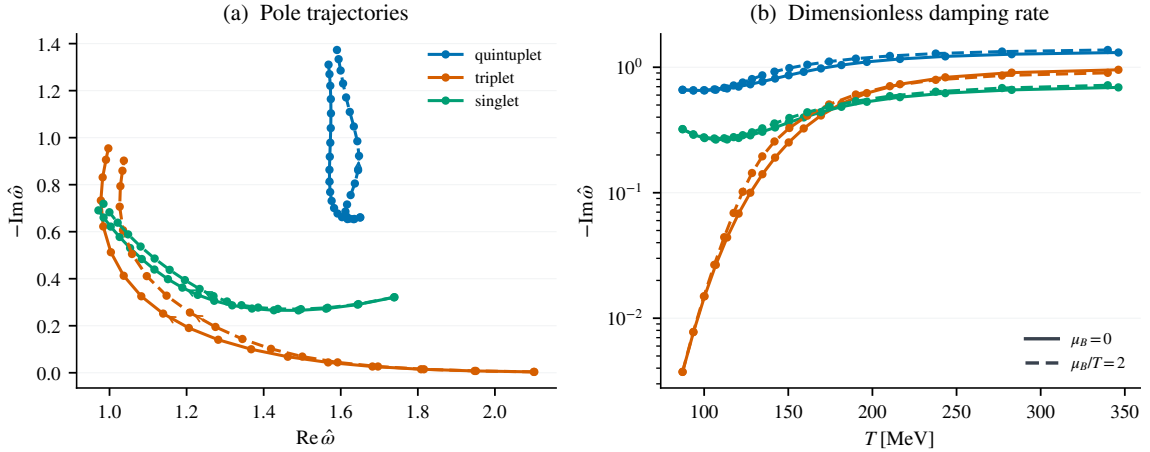


Figure 11. Validated leading homogeneous poles on two controlled thermodynamic trajectories. Arrows point toward higher temperature; solid and dashed curves denote $\mu_B = 0$ and $\mu_B/T = 2$. The triplet becomes long lived toward the low-temperature endpoint while remaining gapped.

D.4 Comparison with the 2018 EMD realization

To separate solver performance from model dependence, we replace the PHA potential, Maxwell coupling, and scale by the functions and parameters of ref. [10] without changing

either numerical algorithm. A local reconstruction gives its endpoint at (89.122, 723.551) MeV, consistent with the rounded (89, 724) MeV quoted there. At this endpoint and at two common neutral states, all nine published relaxation times are reproduced within 0.0111 fm/c; the maximum shooting source residual is 7.5×10^{-9} and the largest spectral/shooting distance is 0.0074 in $\mathbf{w}$.

Table 8. Linear relaxation times in the 2018 EMD realization and the PHA MAP model. “Published” denotes the approximate values quoted in ref. [10]; endpoint rows use each model’s own CEP.

state	model	T [MeV]	μ_B	τ_5	τ_3 [fm/c]	τ_1
high T	2018 published	400	0	0.060	0.080	0.110
	2018 reproduced	400	0	0.059	0.080	0.108
	PHA MAP	400	0	0.059	0.081	0.112
neutral crossover	2018 published	145	0	0.260	1.000	0.700
	2018 reproduced	145	0	0.261	1.008	0.703
	PHA MAP	145	0	0.261	1.025	0.634
model CEP	2018 published	89.0	724.0	0.330	2.150	1.260
	2018 reproduced	89.1	723.6	0.329	2.139	1.252
	PHA MAP	103.8	593.3	0.349	2.934	1.182

At $T = 400$ MeV the two models differ by at most 0.004 fm/c. At 145 MeV the PHA singlet relaxes about ten percent faster. At their respective endpoints both models have $\tau_3 > \tau_1 > \tau_5$, while the PHA triplet time is about 37 percent larger. This is a cross-model response comparison, not a universality or critical-exponent statement. These homogeneous results validate the common background/spectral infrastructure but are not used to infer diffusion or spinodal growth.

E Reproducibility and data products

The complete Route-A analysis is regenerated from the repository root by

```
python -m pip install -r analysis/requirements.txt
python analysis/reproduce_reference_cep.py --lines 800 --neighbors-only \
  --output results/python/reference_cep_neighbors_N800.json
python analysis/locate_reference_cusp.py --phi-step 0.01 \
  --charge-step 0.001 --uv-tolerance 1e-5 \
  --output results/python/reference_cusp_uv1e5_h10.json
python analysis/build_phase_diagram.py --n-phi 141 --n-charge 121 \
  --phi-min 2.2 --phi-max 6.5 --charge-min 0.02 --charge-max 0.62 \
  --mu-min 594 --mu-max 850 --mu-count 65
python analysis/validate_phase_diagram.py
```

```

python analysis/plot_cep_reproduction.py
python analysis/run_numerics.py
python derivations/generate_linearized_emd.py
python derivations/validate_linearized_emd.py
python analysis/validate_decoupled_qnms.py
python analysis/build_homogeneous_trajectories.py
python analysis/compare_2018_emd.py
python analysis/plot_qnm_validation.py
python analysis/plot_homogeneous_trajectories.py
python analysis/validate_coupled_qnms.py
python analysis/run_finite_k_physics.py --stage all
python analysis/run_posterior_uq.py --samples 25 --workers 4
python analysis/plot_posterior_uq.py

```

The posterior driver contains its adaptive cusp fallback and therefore does not require manual sample removal or retuning. The run uses the checksum-recorded source HDF5 and selects sample 74 with $\log \mathcal{L} = 31.0751$. The primary Route-A outputs are `coupled_qnm_validation.csv`, `finite_k_hydrodynamic_dispersion.csv`, `cep_critical_scaling.csv`, `cep_q4_dispersion.csv`, `spinodal_dispersion.csv`, and `posterior_uq_samples.json`, with JSON summaries storing fits and acceptance thresholds. The finite-momentum scan is checkpointed after every root.

The retained validation products include `thermodynamic_validation.csv` and the validated-mode, homogeneous-trajectory, and 2018-benchmark CSV files. Their JSON sidecars store the horizon and UV settings, branch labels, source residuals, condition estimates, tracking parents, eigenfunction overlaps, and independent-method envelopes. Thus the appendices are regenerated data products, not historical prose detached from the current pipeline.

The source HDF5 checksum and exact sample attributes are recorded in the data manifest. Source code and machine-readable outputs are available at <https://github.com/RenatoPhys/pha-qnm>; the upstream posterior data are archived with ref. [9]. The author-controlled identity, funding, arXiv, and persistent-release metadata must be supplied before submission.

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